SOCIOTHON 2026 • NATIONAL SERVICE SCHEME (NSS) IIT ROORKEE

ECOCAMPUS IIT ROORKEE

"Turning IIT Roorkee's 5,500 kg/day Waste Liability into a Self-Funding Circular Economy through 4-Way Ecosystem Incentive Alignment."

STAKEHOLDER 01

Campus Students

AI Camera Sorter, 2-Tap Overflow Report (+35 Pts) & Nescafe Pass

STAKEHOLDER 02

Sanitation Field Crew

Hindi Voice AI, Anti-Spam Photo Verification & 30% UPI Bonus

STAKEHOLDER 03

Estate Command

200-Bin Load/Ultrasonic Telemetry, 2-Opt TSP & B2B Auctions

STAKEHOLDER 04

B2B Recyclers (ITC)

Live Reverse Bidding on Grade-A Baled Lots & Digital Gate Passes

Team Name: **Team EcoCampus** | Pilot Scope: **Govind Bhawan (200 Bins) & IITR Campus (365 Acres)** | github.com/NIKHILESH-TS/EcoCampus-IITR

EMPIRICAL CAMPUSBASELINE & ROOT CAUSE

THE PROBLEM — WHAT CHALLENGE ARE WE SOLVING?

Grounded in the IIT Roorkee Empirical Solid Waste Audit (MDPI Study: 5,500 kg/day across 12,000 Residents)

01 // EMPIRICAL IIT ROORKEE WASTE STREAM (5.5 METRIC TONNES / DAY)

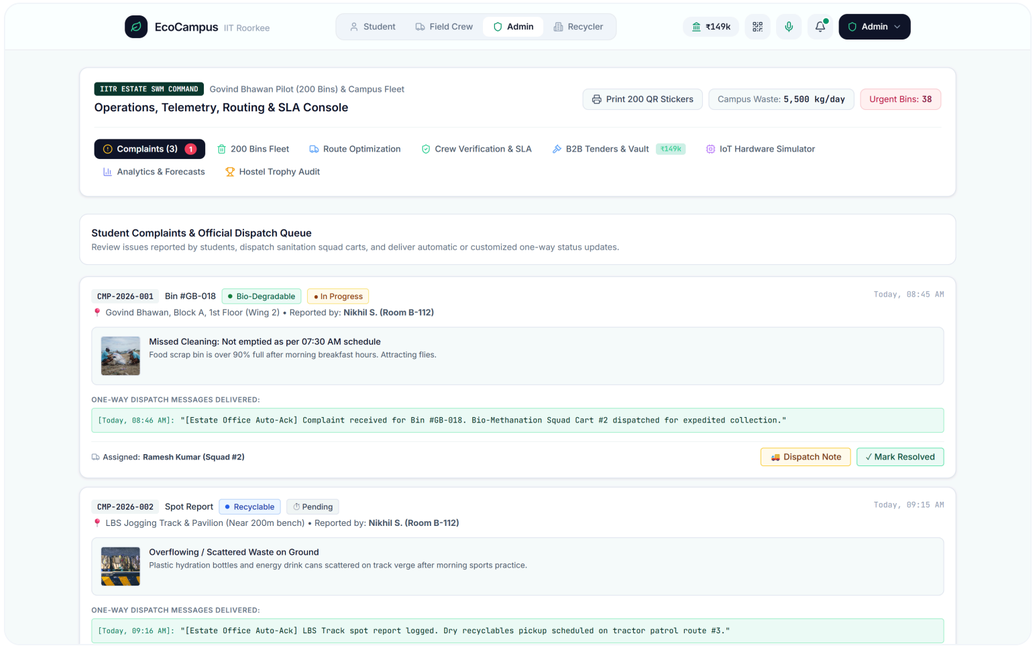
83.5% of IITR's Waste is Recoverable—Yet Value is Lost at the Bin

- **Organic & Mess Food (30.5% — 1,676 kg/day):** Contaminates dry paper bins when unsegregated at hostel wings.
- **Paper & Cardboard (28.7% — 1,580 kg/day):** High-value Kraft fiber across LHC, Library & 14 Bhawans.
- **Plastics & Polymers (11.1% — 613 kg/day) + Metals (3.2% — 175 kg/day):** Sold at flat informal scrap rates.
- **Garden Foliage (10.3% — 569 kg/day) & Inert (10.8% — 595 kg/day):** 200 bins in Govind Bhawan lack real-time fill or weight sensors.

02 // THE CORE FAILURE: BROKEN INCENTIVES

Why Traditional Smart-Bin & Complaint Apps Fail

- **Students Have Zero Skin in the Game:** No tangible reward for source segregation; open complaint forms get spammed with false alarms or ignored completely.
- **Sanitation Workers Are Disconnected:** Fixed wages regardless of segregation quality, English-only interfaces, and blind tractor routes burning **1,420 L/month** excess diesel.
- **60% Value Destruction in B2B Scrap:** Mixed/wet recyclables fetch only **₹8/kg** from local aggregators instead of **₹17.20/kg** from corporate mills (ITC / Nepra).



Live Prototype Screenshot — Estate Command Console Monitoring 200 Govind Bhawan Bins & Dispatch SLA

THE SOLUTION ARCHITECTURE

THE IDEA — ECOCAMPUS IITR CIRCULAR OS

A Closed-Loop 4-Stakeholder Ecosystem Connecting Source Segregation, Field Verification & B2B Scrap Auctions

STEP 01 // STUDENT

AI Scan & 2-Tap Report

Students classify waste via MobileNetV3 camera/voice (**+10 Pts**) & report full bins via QR (**+35 Pts** on verify / **-25 Pts** false alarm).

STEP 02 // FIELD CREW

Hindi Voice & Proof

Squad #2 receives Hindi ElevenLabs audio alerts, uploads ground-truth photo proof, sorts secondary stream & earns **+₹150 UPI bonus**.

STEP 03 // ESTATE ADMIN

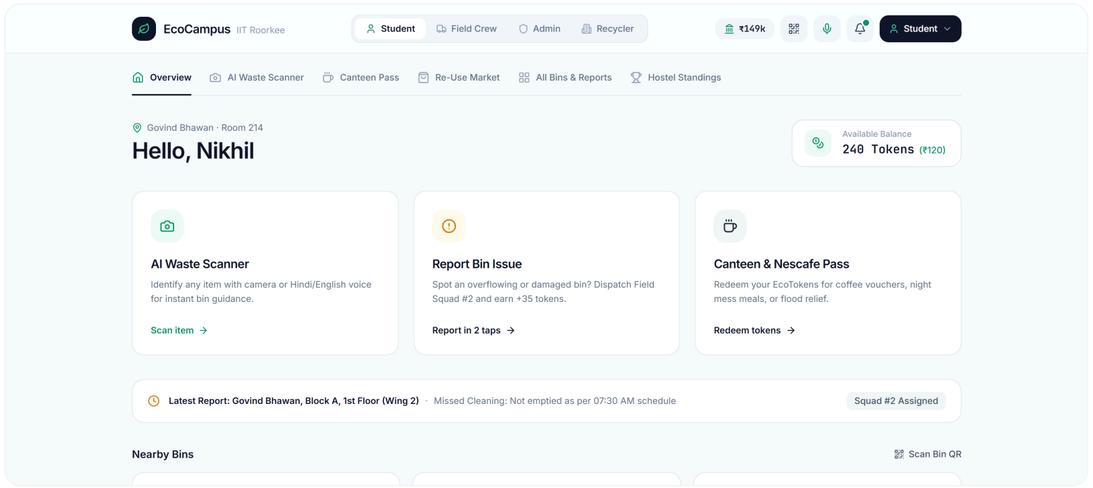
IoT Fusion & Baling

ESP32 Ultrasonic + HX711 Load Cells trigger 2-Opt TSP tractor routes; Central SWM Yard bales weekly dry recyclable lots.

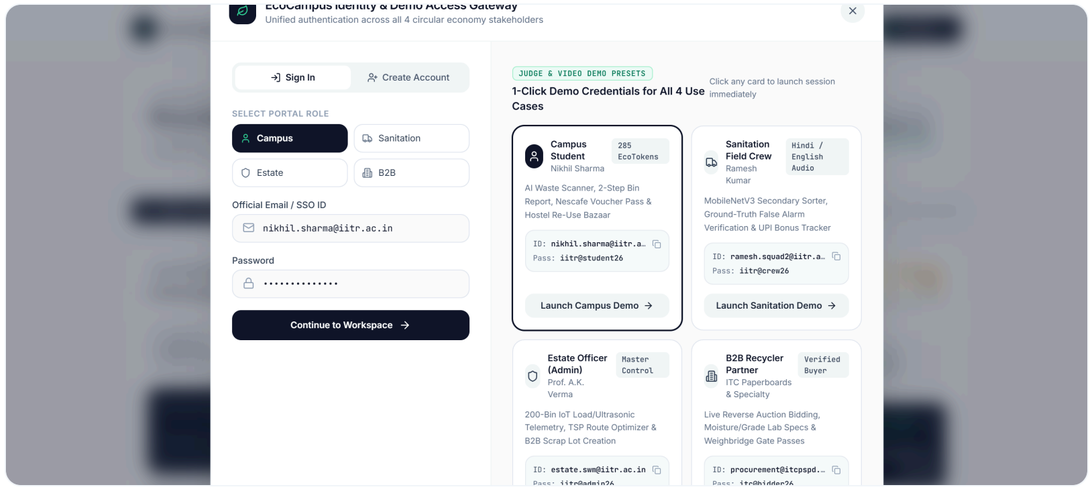
STEP 04 // B2B & VAULT

Reverse Auction Vault

ITC Paperboards & Nepra bid live on baled lots; proceeds flow into the **Campus Circular Vault** to fund Nescafe & Crew UPI.



Live Student Overview — Calm Apple/Linear UI with 3 Core Actions & Nearby Bin Telemetry

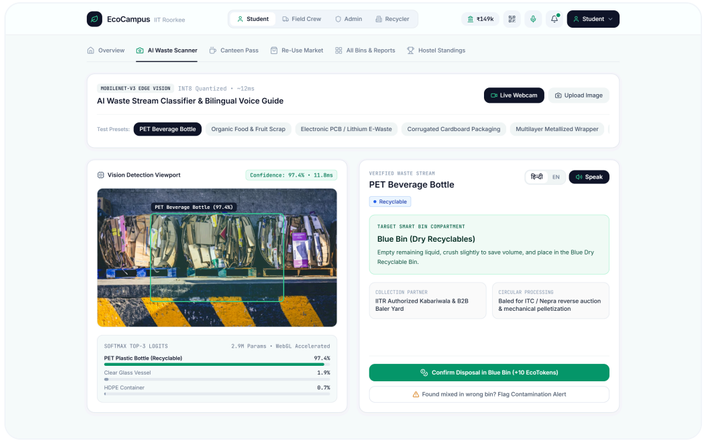


Live 4-Persona SSO Gateway — Pre-Configured 1-Click Demo Credentials for All 4 Stakeholders

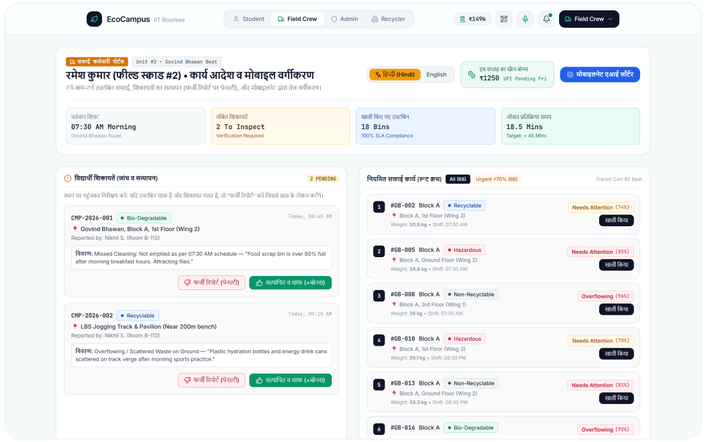
MULTI-PERSONA PRODUCTION SUITE

KEY FEATURES — BUILT & WORKING END-TO-END

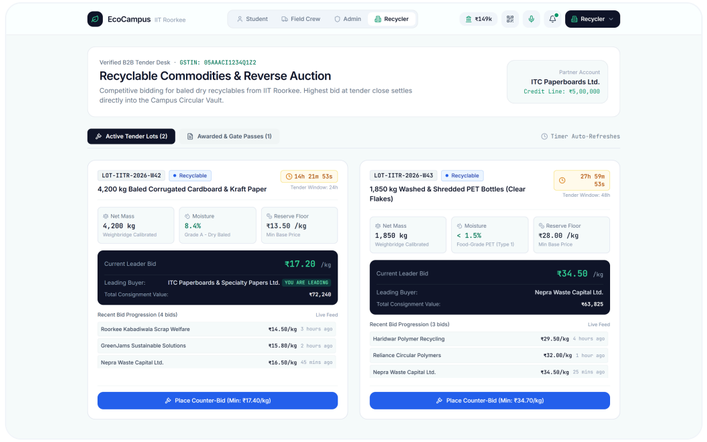
Tailored Interfaces for Students, Hindi-Speaking Sanitation Crew, Estate Officers & Corporate B2B Recyclers



1. Student AI Vision Scanner (Live Webcam + Softmax Logits + Bilingual Audio)



2. Field Crew Hindi Portal (Ground-Truth Dispute Proof & MobileNet Sorter)



3. B2B Recycler Reverse Auction Desk (ITC Live Bidding & QR Gate Passes)

AI & Voice Guidance

Live WebRTC camera + MobileNetV3 classification with ElevenLabs Hindi/English spoken disposal instructions.

Anti-Spam SLA Verification

Workers inspect reported bins and upload timestamped proof: legitimate reports earn +35 Pts; fake reports deduct -25 Pts.

Hostel Re-Use Bazaar

Peer-to-peer zero-waste exchange for academic books, kettles, drafters & skates priced algorithmically in EcoTokens.

B2B Tender & Gate Passes

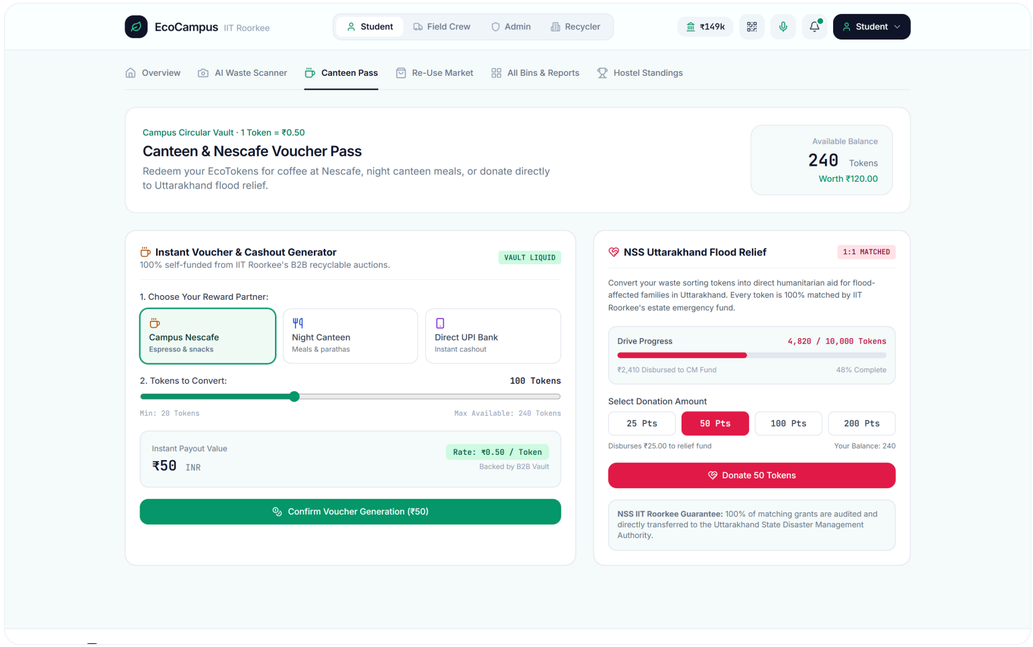
Moisture-certified (<8%) baled lots with quick counter-bid buttons (+₹0.20 to +₹2.00/kg) and automated Weighbridge PDF passes.

CORE INNOVATION & GAME THEORY

THE DIFFERENTIATOR — 4-WAY INCENTIVE ALIGNMENT

We Replaced Voluntary Charity with a Self-Enforcing Economic Flywheel Backed 100% by B2B Scrap Surplus

ACTOR	INCENTIVE / REWARD	ANTI-GRIEFING / QUALITY GUARDRAIL	VAULT BACKING
1. Students Govind Bhawan	**+10 Pts** (AI Scan) **+35 Pts** (Verified Report)	**-25 Token Penalty** if Field Crew photo proves False Alarm	**30% of Vault** 1 Pt = ₹0.50 at Nescafe & Mess
2. Field Crew Sanitation Squads	**+₹150 Bonus** per verified SLA fix + weekly share	Mandatory timestamped after-photo & MobileNet secondary sort	**30% of Vault** Direct Friday UPI Payout
3. B2B Buyers ITC / Nepra	Uncontaminated Grade-A baled paper & PET (<8% moisture)	Reserve floor price & Century Gate weighbridge tare audit	Pays **₹17.20/kg** (+56% over informal rate)
4. IITR Admin Estate & NSS	Zero-capex self-maintaining IoT fleet & social impact	Automated smart-contract partition of every auction settlement	**25%** Hardware Upkeep **15%** NSS Flood Relief



Live Student Canteen Pass & NSS Uttarakhand Flood Relief Matching Pool (100% Backed by ₹149k Campus Vault)

GAME-THEORETIC NASH EQUILIBRIUM AGAINST GRIEFING

Expected Utility of Truthful Reporting **E[U(Truth)] = +35 Tokens (+₹17.50)** vs. Expected Utility of Spamming **E[U(Spam)] = -25 Tokens (-₹12.50)**. Trolling becomes economically irrational while genuine reporting is immediately rewarded.

ENGINEERING & ALGORITHMIC ARCHITECTURE

TECH STACK & CORE ALGORITHMS

Edge Neural Vision, Dual-Sensor Physical Heuristics, Multilingual Voice Synthesis & Combinatorial Optimization

01 // EDGE VISION

MobileNetV3-Small (INT8)

Depthwise separable convolutions + Hard-Swish activation. **2.9M params, 2.5 MB weights, ~12ms WebGL latency**. Computes Top-3 Softmax logits across 6 waste streams.

```
h-swish(x) = x · ReLU6(x + 3) / 6
```

02 // IOT SENSOR FUSION

Apparent Density Heuristic

Combines **HC-SR04 Ultrasonic** (fill volume ΔV) + **HX711 50kg Load Cell** (mass Δm) on an ESP32 node (**₹465/bin BOM**) to detect wet food in dry paper bins.

```
ρ = Δm / ΔV > 0.65 kg/L ⇒ CONTAMINATION
```

03 // BILINGUAL VOICE AI

ElevenLabs Multilingual v2

Low-latency REST streaming + Web Speech fallback delivering natural **Hindi & English** spoken instructions for sanitation crews and hostel students.

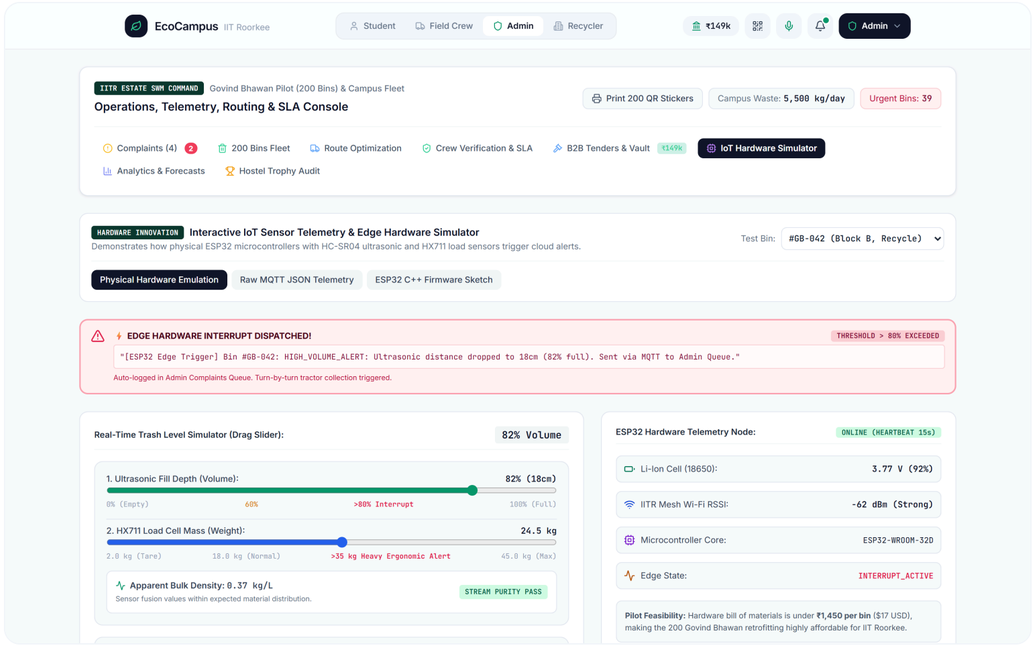
```
Model: eleven_multilingual_v2 (HI / EN)
```

04 // ROUTING & PRICING

2-Opt TSP & Re-Use Algo

2-Opt edge-swapping TSP cuts tractor path by **34%**, visiting only bins where Fill ≥ 80% or Mass ≥ 35 kg. Dynamic P2P Bazaar valuation rewards reuse.

```
P = (B_cat × M_cond) + U_acad + A_free
```



Live ESP32 Ultrasonic + HX711 Load Cell Hardware Telemetry Simulator with Real-Time Bulk Density Anomaly Detection

QUANTIFIED OUTCOMES & ROLLOUT ROADMAP

IMPACT & SCALABILITY

Unit Economics & Phased Expansion from Govind Bhawan (200 Bins) to Pan-India Smart Campuses

73.4%

Landfill Diversion

4,037 kg/day diverted from Roorkee municipal dump into biogas & B2B recycling.

₹2.4L/mo

B2B Scrap Surplus

Unlocked from 1,580 kg/day paper & 613 kg/day plastic via ITC/Nepra reverse bidding.

384 T/yr

CO_2e Avoided

Plus 1,420 Litres/month tractor diesel saved via sensor-triggered 2-Opt TSP routing.

18 Days

Hardware Payback

₹465 retrofittable ESP32 bin node recovered in under 3 weeks of Grade-A scrap premium.

PHASE 01 // MONTHS 1-2

Govind Bhawan Pilot (200 Bins)

Deploy waterproof QR stickers on all 200 Govind Bhawan bins and ₹465 ESP32 sensor nodes on 25 high-traffic wing bins. Onboard Sanitation Squad #2 and activate Campus Nescafe voucher redemption.

CAPEX: ₹11,625 • Self-Funded in Month 1

PHASE 02 // MONTHS 3-6

Full IIT Roorkee Rollout (365 Acres)

Scale across all 14 Bhawans, Lecture Hall Complex (LHC), Central Library, and Faculty Quarters (capturing all 5,500 kg/day). Integrate Central SWM Baler Yard weighbridge for weekly ITC/Nepra reverse tenders.

12,000 Residents • ₹28.8 Lakhs/Yr Gross Yield

PHASE 03 // YEAR 1+

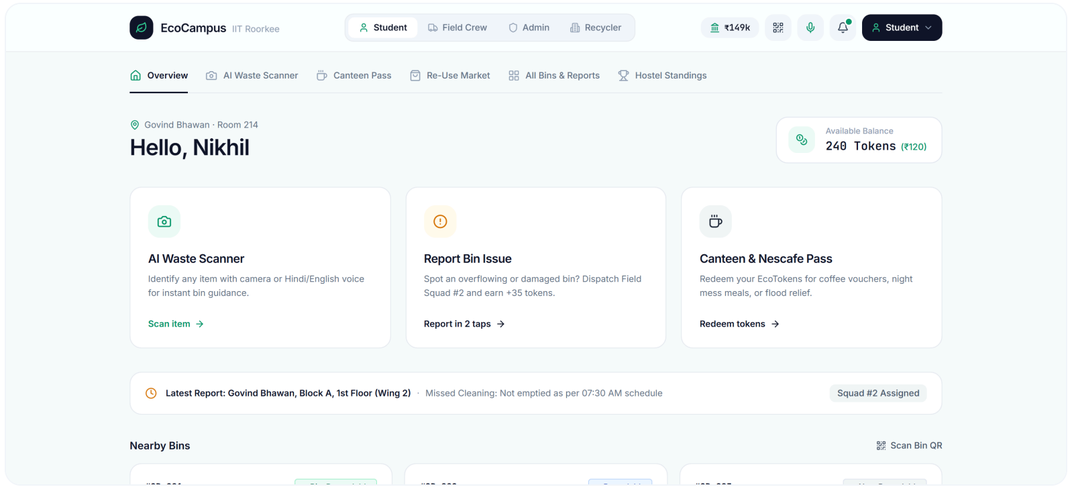
Pan-IIT & Smart Township SaaS

Replicable plug-and-play circular treasury blueprint for 23 IITs, 31 NITs, AIIMS campuses, and RWA townships—turning institutional waste management from a municipal cost sink into a revenue-positive asset.

Zero-Capex SaaS + EPR Credit Monetization

FROM IDEA TO IMPACT • ONE POWERFUL STATEMENT

“Waste is only a design flaw of broken incentives. When a student’s clean segregation buys their evening Nescafe, a worker’s verified cleanup funds their Friday UPI bonus, and ITC bids live for campus scrap—zero-waste becomes unstoppable.”



Live Production Prototype — github.com/NIKHILESH-TS/EcoCampus-IITR



THANK YOU • READY FOR LIVE DEMO & Q&A

Repository & Live Demo: https://github.com/NIKHILESH-TS/EcoCampus-IITR